Predicting Customer Churn Using Binary Classification Models

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Abstract

Customer churn refers to the process of customers leaving a service or discontinuing their relationship with a business. This study develops both a traditional machine learning classifier and a deep learning model to predict customer churn based on past customer data, including demographic details, insurance history, and engagement metrics. Prior to model training, the dataset was preprocessed by standardizing numeric features and using one-hot encoding for categorical features. Our machine learning classifier, implemented with Scikit-learn, achieves a mean cross-validation score of 90.17% on test data. For our deep learning approach, we employ PyTorch and structure the data into tensors for efficient computation. After 50 epochs with a batch size of 32, our deep learning model achieves an average predictive accuracy of approximately 91%. These results highlight the effectiveness of both machine learning and deep learning approaches in predicting customer churn, providing valuable insights for financial institutions aiming to improve customer retention strategies.

1 Introduction & Motivation

Customer churn, the phenomenon of customers leaving a service or discontinuing their relationship with a business, presents a significant challenge to companies in highly competitive markets. Predicting customer churn is a crucial step toward implementing retention strategies and improving customer satisfaction. In banking specifically, churn rates can reach up to 25%, with a large portion of attrition occurring within the first 90 days of a new account (Egan, 2023). This customer churn can lead to substantial financial losses, particularly when valuable or long-term customers are lost. In fact, acquiring a new customer can cost five to seven times more than retaining an existing one (Kumar, 2022). It is estimated that increasing customer retention by just 5% can boost profits by 25% to 95% (Bain&Company, 2022). Therefore, leveraging modern deep learning techniques to predict churn accurately can empower organizations to take proactive measures aimed at retaining their customer base. This project aims to build a robust classification model that can accurately predict customer churn, offering valuable insights to stakeholders and enhancing decision-making processes.

2 Methodology

The dataset used in this project was derived from real bank data from 2022, slightly modified to preserve privacy and enhance analytical richness. The dataset consists of 24000 customers, each with features describing customer demographics, transaction history, account activity, and other relevant attributes. Preprocessing steps included handling missing data, normalizing numerical features, and encoding categorical variables as binary to prepare the data for modeling. In our attempts to efficiently predict customer churn, we trained and tested the accuracy of two binary classification models. The first approach is a traditional machine learning algorithm known as the random forest classifier model, and the second is a deep learning approach utilizing a fully connected feed-forward neural network.

2.1 Machine Learning Approach – The Random Forest Classifier Model

The Random Forest model is a widely used ensemble learning technique primarily employed for classification and regression tasks. It operates by constructing a “forest” of decision trees during the

training phase and outputting the majority classification of all the individual trees. Introduced by Leo Breiman, Random Forests address the high variance commonly observed in individual decision trees by combining their outputs, thereby significantly improving predictive accuracy and controlling overfitting (Breiman, 2001).

Each tree in the forest is trained on a bootstrap sample of the data. Bootstrapping is a sampling method where many samples are drawn with replacement to approximate the variability of the population. Furthermore, at each node split during the tree construction, a random subset of features is considered, rather than evaluating all available features. This injection of randomness not only de-correlates the trees but also enhances model robustness by ensuring that no single predictor dominates the ensemble's behavior.

This model is resilient to noise, capable of handling high-dimensional data, and can provide intrinsic measures of feature importance. In practice, the Random Forest model proved to be an effective means of predicting customer churn. The Random Forest's ease of use, minimal hyperparameter tuning requirements, and strong empirical results have made them a foundational model in both academic research and industry applications. This reputation made the Random Forest our first method choice to predict customer churn.

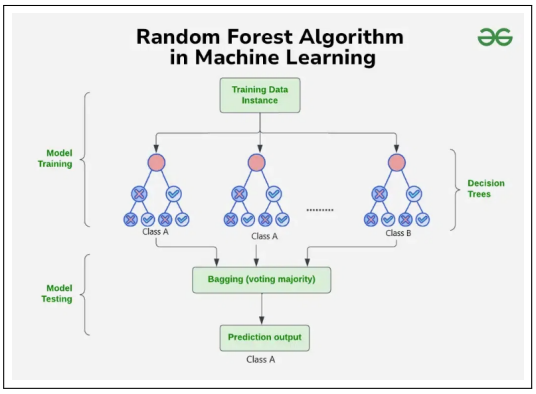

Figure 1: Random Forest Algorithm.

2.2 Deep Learning Approach – Feed Forward Neural Network

For the deep learning approach, we constructed a neural network consisting of several fully connected layers. This architecture contains trainable linear layers, reducing neuron count from 64 at the input layer to 2 at the output layer. Each linear layer is followed by hidden layers utilizing ReLU activation functions, which mitigate the risk of encountering the vanishing gradient problem and introduce non-linearity to the model, allowing it to learn more complex patterns in our data. To improve the model's generalization capabilities and prevent overfitting, we applied dropout regularization after each ReLU function. Dropout randomly deactivates a fraction of neurons during training, forcing the network to learn more robust features (Nitish Srivastava, 2014). More specifically, our neural network consists of seven layers in total, including three trainable linear layers, ReLU activation functions after the first two linear layers, and a dropout rate of 0.3 (30% of neurons deactivated) applied after each ReLU function.[1]

The model was trained over the course of 50 epochs with a batch size of 32 using the Adam optimizer, a variant of stochastic gradient descent known for its computational efficiency and low memory requirements (Diederik P. Kingma, 2015). The loss function employed was binary cross-entropy, which measures the dissimilarity between the true labels in a binary classification (0 or 1) and the predicted logits or probabilities, and penalizes the model for incorrect predictions, with higher penalties for more confident wrong predictions. This made binary cross-entropy loss ideal for the binary classification task of predicting whether a customer would churn or not.

[1] See usage of `torch.nn.Linear()`, `torch.nn.ReLU()`, and `torch.nn.Dropout(0.3)` in AI_Models.ipynb section 5.3

3 Results

Comparative analysis between traditional machine learning models and the deep learning model demonstrated that the latter achieved superior predictive performance. However, the deep learning model required more computational resources and training time. The inclusion of dropout regularization significantly contributed to the model's robustness by mitigating overfitting. Overall, our methodology underscores the utility of combining traditional machine learning models with modern deep learning techniques to achieve accurate and reliable predictions. The following sections will delve into the results obtained from these models and discuss their implications for customer churn prediction.

3.1 Statistical Analysis

To assess both of the classification models, we used evaluation metrics rooted in precision and recall. Precision represents the proportion of true positive classifications out of all the positive classifications that the model made. Precision can also be understood as: "when the model says yes, how often is it right?" On the other hand, recall represents the proportion of true positive classifications the model established out of all the true positive classifications in the data. Recall can also be understood as: "of all the true yes cases, how many did the model find?"

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}} \qquad \text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}}$$

The F1 score is a metric that balances precision and recall into a single number, especially useful when classes are imbalanced or when false positives and false negatives are both important. The equation for F1 score uses the harmonic mean rather than arithemetic mean. The harmonic mean penalizes extreme values more than the arithmetic mean. So if either precision or recall is low, the F1 score will also be low — it reflects the trade-off between precision and recall in a single value. The Random Forest model achieved a weighted average F1 score of 90%.

$$F_1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

We ran a precision and recall test on the deep learning model to see the relationship between type 1 and 2 error (false positives and false negatives). In the precision-recall curve below (Figure 1), there is not a dramatic tradeoff between precision and recall. This means that our training set is well balanced and good for all 4 binary cases (between the predicted classification and the actual classification). The confusion matrix below demonstrates this, as the bottom left (type II error / false negative) and upper right (type I error / false positive) boxes hold the lowest values.

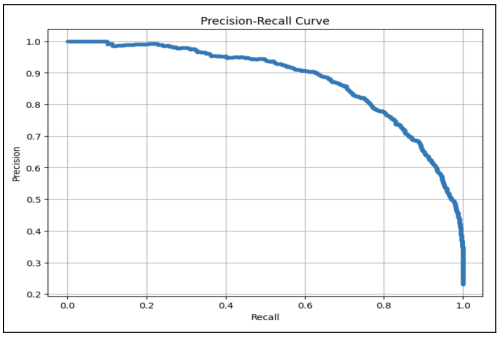

Figure 2: Precision recall curve.

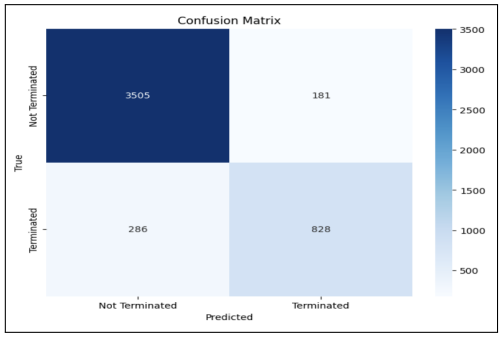

Figure 3: Confusion matrix.

3.2 Implications

Implementing a 90%–accurate churn-prediction model enables banks to intervene proactively with high-risk customers, tailoring retention offers such as fee waivers or rate discounts to those most

likely to leave. Over time, this makes for personalized engagement strategies. However, maintaining performance requires continuous monitoring for model drift as customer behavior and market conditions change, alongside regular retraining. Banks must also address ethical concerns by ensuring that predictive factors do not introduce demographic bias. Embedding churn signals into daily operations can drive a broader culture of data-driven decision making, while successful retention creates an area for cross-selling loans, cards, and investment products.

For a mid-sized bank with one million retail accounts and a 20% annual churn rate (200,000 customers), a model recalling 90% of churners flags roughly 200,000 accounts (including 10% false positives). At $50 per targeted intervention, retention efforts cost $10 million. If each retained customer yields $3,500 in lifetime value, saving 180,000 true positives generates $630 million in gross benefit, netting approximately $620 million after incentive costs (even accounting for development and maintenance expenses under $3 million). Crucially, the bank breaks even by saving as few as 18,000 customers (10% of whom are flagged), showing that modest improvements in churn reduction translate into tens of millions in net gains.

3.3 Future Improvements

To improve the deep learning model's ability to predict churn, we will focus on making our data more robust and comprehensive. This includes creating new features like the ratio of a customer's balance to their tenure (e.g., "balance per year"), customer activity frequency, or engagement scores combining several behaviors. Incorporating external datasets such as regional economic indicators (unemployment rates, inflation) or customer service interaction logs could provide more context around why a customer might leave.

The current model can be further improved by applying hyperparameter tuning (using methods like randomized search, grid search, or Bayesian optimization) to find the best learning rates, dropout rates, hidden layer sizes, and batch sizes. Trying more advanced architectures like attention-based models or ensemble methods that combine deep learning with tree-based models (e.g., TabNet, DeepGBM) could significantly boost performance, especially with richer features.

Finally, addressing class imbalance more aggressively through SMOTE (Synthetic Minority Oversampling) or focal loss would help ensure the model doesn't become biased toward predicting non-churners, leading to better recall for the critical "likely to churn" customers.

4 Discussion

4.1 Principal Findings

Our experiments on both models show that the deep learning model achieved 91.14%, outperforming Random Forest (one of the best traditional machine learning algorithms) by approx. 1%. Although this gain is low per average comparison, it shows that even on a structured, low-dimensional banking dataset, there are weak, non-linear signals that traditional algorithms do not fully exploit.

In our training of both models, most features are numerical or one-hot encoded categorical variables that were already amenable to traditional machine learning. Due to the small nature of the feature space, hand-crafted feature engineering plus the tree-based model used can find the right curve nearly as well as any 3-7 layer neural network. The 1 percentage point lift we observe thus captures the marginal value of automatically learning higher-order interactions, which is useful but not transformative for this data regime.

4.2 When Deep Learning Could Matter More

If banks can ingest raw, unstructured sources of data, such as call-centre transcripts, customer service emails, or even voice notes of the brief variety, neural networks will only draw further ahead. Modern language and audio transformers are excellent at capturing complex patterns in text sentiments, acoustic tone, or conversation dynamics that are not capturable by traditional ML without dense, domain-specific feature pipelines. For instance, tone-of-voice embeddings from calls will capture frustration signals weeks before transaction behavior shifts.

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